

GOKHAN BALIK, Ph.D.

Cybersecurity & AI Researcher | Systems Developer

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Phone: _____ | Location: St. Louis, MO

PROFESSIONAL SUMMARY

Cybersecurity and AI researcher and systems developer specializing in geospatial cyber risk intelligence, critical infrastructure resilience, and trustworthy AI-assisted decision support. Work integrates cybersecurity, artificial intelligence, geographic information systems (GIS), and risk assessment to analyze how cyber risks emerge, propagate, and affect interconnected infrastructure systems.

Developer of a coherent body of independent work in infrastructure resilience, cyber risk assessment, and explainable decision support, including a Scopus-indexed publication on GIS-enabled cybersecurity and four completed platforms: CyberGRIP, a geospatial cyber risk and resilience intelligence platform; an AI-assisted network discovery and security assessment tool; a NIST-aligned AI-assisted risk assessment framework; and a document-grounded AI framework for trustworthy decision support.

Builds on a Ph.D. in GIS-based risk modeling, 14+ years of academic research (30 publications, 90 citations, 6 funded projects), and ongoing graduate study in Cybersecurity with an AI emphasis, advancing an interdisciplinary endeavor in AI-driven geospatial cyber risk intelligence, infrastructure interdependency analysis, and resilience-informed decision-making for critical infrastructure protection.

AREAS OF EXPERTISE

- Geospatial Cyber Risk Intelligence (GIS & Cybersecurity)
- Critical Infrastructure Protection, Risk & Resilience
- AI-Assisted Cyber Risk Assessment & Decision Support
- Infrastructure Dependency & Network Analysis
- Cascading Failure & Complex Systems Risk Modeling
- Trustworthy & Explainable AI (XAI) and Grounded RAG Systems
- Cyber Risk Intelligence & Resilience-Informed Decision-Making

EDUCATION

- **M.S. in Cybersecurity (AI Emphasis)** | Webster University, St. Louis, MO | Aug 2025 – May 2027 | GPA: 3.9 | Scholarship Recipient | Leadership: Cybersecurity CTF Team Leader
- **A.S. in Computer Programming** | Anadolu University, Remote | Sep 2024 – Feb 2027
- **Ph.D. in Landscape Architecture** | Ege University, Turkey | Jul 2011 – Mar 2018 | Dissertation: *Creating a GIS based Ecological Risk Assessment Model*
- **M.S. in Landscape Architecture** | Ege University, Turkey | Sep 2006 – Jan 2010
- **B.S. in Landscape Architecture** | Ege University, Turkey | Sep 2001 – Aug 2006

WORK EXPERIENCE

Cybersecurity & AI Researcher and Systems Developer | Aug 2025 – Present

Independent Research & Development, St. Louis, MO | Part-Time

- Independent research and systems development at the intersection of cybersecurity, AI, critical infrastructure, cyber risk assessment, GIS, NLP, computer vision, and intelligent decision-support systems.

Ethical Hacker, Programmer, and IT Support Technician | Aug 2024 – Jul 2025

StallionDC Ltd., U.K. | Part-Time, Remote

- Performed penetration testing and vulnerability assessment to identify security weaknesses and documented mitigation guidance to strengthen data center security.
- Developed automation apps for Linux and network support workflows, improving operational efficiency by **25%** by reducing repetitive manual tasks.

Computer Vision Engineer | Oct 2023 – Feb 2024

Seattle-based Tech Company, Seattle, WA | Internship, Remote

- Developed a YOLO based object detection model for satellite and UAV imagery to detect small objects under variable atmospheric conditions.
- Built a real time preprocessing pipeline for images, videos, and live streams, applying normalization and quality enhancement and achieved over **90%** precision.
- Developed two wind turbine crack detection workflows using UAV infrared and RGB imagery, using YOLO for detection and Mask R-CNN as an alternative modeling and segmentation approach.

Associate Professor, Researcher and Lecturer | Feb 2012 – Jul 2025

Trakya University & Ege University, Turkey | Full-Time

- Co-led an AI based medical diagnostics project, publication in press.
- Contributed to funded interdisciplinary research projects in GIS, risk assessment, spatial analysis, environmental modeling, experimenting and applied technology.
- **6** funded projects, **30** publications, **3** awards, **2** interviews, **90** citations, and h-index **5**.
- Developed a student registration application, reducing processing time by **80%**.
- Organized national and international symposiums and workshops.

Student Food Service Worker | Aug 2026 – Present

American Dining Creations, St. Louis, MO | Part-Time

- On-campus employment while pursuing graduate studies in Cybersecurity with AI Emphasis.

INDEPENDENT RESEARCH & ENGINEERING PROJECTS

Project 1 – AI Geospatial CIKR Network Risk and Resilience Intelligence Platform:

CyberGRIP is an AI-assisted platform for identifying, analyzing, and prioritizing cyber and resilience risks across critical infrastructure systems by integrating geospatial intelligence, network science, and risk modeling to support resilience planning, mitigation prioritization, and resource-allocation decisions.

- Developed an AI-assisted platform that integrates GIS, network science, risk assessment, and resilience analytics for critical infrastructure systems.
- Modeled infrastructure interdependencies and cascading-risk pathways to support analysis of critical nodes, network resilience, and disruption impacts.
- Applied deterministic risk scoring, geospatial intelligence, and grounded AI-assisted decision support to improve transparency and risk-informed decision-making.

Project 2 – AI-Assisted Network Discovery and Security Assessment Tool:

An AI-assisted platform for discovering, classifying, and mapping network assets by combining automated asset discovery, topology analysis, and security assessment to improve network visibility, identify security risks, and support dependency-aware cybersecurity planning.

- Developed an AI-assisted platform for network asset discovery, topology mapping, and cybersecurity assessment.
- Enhanced infrastructure visibility through automated asset identification, device classification, and dependency-aware analysis.
- Applied AI-assisted risk analysis to support security assessment and decision-making in complex networks.
- Served as the foundation for later work in cyber risk intelligence, infrastructure interdependencies, and critical infrastructure resilience.

Project 3 – AI Assisted Risk Assessment Framework (NIST aligned):

An AI-assisted framework for identifying, analyzing, and prioritizing cybersecurity risks across organizational environments by integrating deterministic risk scoring, control-gap analysis, and NIST-informed guidance to support mitigation planning, resilience improvement, and risk-informed decision-making.

- Developed a NIST-informed framework for cyber risk assessment, control-gap analysis, and mitigation prioritization.
- Integrated deterministic risk scoring with AI-assisted decision support to improve transparency, traceability, and risk-informed decision-making.
- Applied authoritative NIST guidance to support cybersecurity resilience, organizational risk management, and executive decision-making.

Project 4 – Document Grounded AI Assistant and RAG Framework:

An AI-assisted framework for generating trustworthy, evidence-based insights from organizational documents by integrating retrieval, source attribution, and faithfulness verification to support explainable decision-making, risk analysis, and compliance workflows.

- Developed a document-grounded AI framework integrating retrieval augmented generation, evidence attribution, and faithfulness verification.
- Improved the transparency, traceability, and reliability of AI-assisted outputs through source-grounded retrieval and validation mechanisms.
- Designed evidence-based decision-support capabilities for cybersecurity, risk assessment, and compliance-oriented workflows.

SELECTED PUBLICATIONS

- Balik, G. (2026). *Integrating GIS and Cybersecurity: Spatial Intelligence for Critical Infrastructure Risk and Resilience*. GIS Odyssey Journal, 6(1), 45–62. <https://doi.org/10.57599/gisoj.2026.6.1.45> (Scopus-indexed).
- Balik, G., et al. (2026). *Privacy-Preserving Decision Support for Emergency Department Triage: A Simulation- Based Benchmarking of On-Premises Retrieval-Augmented and Cloud-Based Large Language Models*. Submitted to an SCI-indexed Springer Nature journal; peer review completed; awaiting editorial reassignment.

RESEARCH IMPACT & PROFESSIONAL SERVICE

- **30** scholarly publications; **90** citations; h-index **5**
- Principal investigator or contributor on **six** funded research projects
- Reviewer for **two** AI research projects funded by the Scientific and Technological Research Council of Turkey (TÜBİTAK)
- **3** research and academic awards; **2** media interviews on research work
- Organizer of national and international academic workshops and symposiums

TECHNICAL SKILLS

- Cybersecurity Domains: AI Security, Application Security, Network Security, Vulnerability Management, Cyber Risk Assessment, NIST Frameworks, OWASP
- Cybersecurity Tools & Technologies: Burp Suite, Wireshark, Metasploit, Nmap, Kali Linux, Virtual Machines (VMs), Open-Source Security Tools
- AI, Machine Learning & Deep Learning: Large Language Models (LLMs), NLP, Retrieval-Augmented Generation (RAG), Computer Vision, Machine Learning, Deep Learning, PyTorch, TensorFlow, Scikit-learn
- Development & Engineering: Python, Bash, SQL, PHP, HTML, CSS, JavaScript, VBA, C#, GitHub, Anaconda, Docker

CERTIFICATES – BOOTCAMPS

- Complete Cybersecurity Bootcamp | Zero to Mastery Academy (ZTM) | Nov 2024
- Complete Ethical Hacking Bootcamp | ZTM | Feb 2025
- Web Security & Bug Bounty: Learn Penetration Testing | ZTM | Feb 2025
- THM (TryHackMe) Courses / Certificates | Feb 2025 – Ongoing
- Complete A.I., Machine Learning and Data Science | ZTM (Oct 2023)
- Complete Python Developer | ZTM | Sep 2023
- Statistics with Python | ZTM | Jun 2024